Technical Specification

Item	Detail
Project	Roost, a verified student sublease marketplace for UC
Course	CS5001 Senior Design I, Assignment 9 (Week 9)
Team members	Danvanth Vanganuri, Jovan Jose Asker Fredy, Mohamed Aidja
Document version	2.0 (2026-10-28)
Traces to	Requirements_Document.md **v2.0 (2026-10-28)**. All REQ, NFR, CON and glossary terms have the meanings given there.
Describes	the Roost prototype, roost-uc-sublease commit `71d351b` (2026-10-26), plus the designed-but-unbuilt items marked **Designed**.

1. Version and Change Log

Version	Date	Change	Author
1.0	2026-10-28	First issue, consolidated from D0, D1 and D2 against RD v1.0 (Node/Express backend, outbox, Render). Not submitted.	Jovan Jose Asker Fredy
2.0	2026-10-28	Rewritten to describe the prototype that the team built and the advisor reviewed: Next.js on Vercel, Supabase (Auth, Postgres with RLS, Realtime, Storage, pg_cron), an Expo phone app. Traces to RD v2.0. The D0 to D2 components are mapped to the prototype in §2.1. Fair Housing screening is re-hosted from the C4 module to a database trigger (§3.6). All design changes since the Week 7 review are logged in §7.	Jovan Jose Asker Fredy
2.1	*pending*	Edits from the advisor review of these drafts (§8).	*TBD*

2. Architecture Summary

Roost is a **client-server web application on a backend-as-a-service**. There is no custom application server. Components:

- **Web client:** Next.js 16 (App Router, React 19, TypeScript, Tailwind), hosted on **Vercel**. It talks directly to **Supabase** with the public publishable key and the user's session token.
- **Supabase** holds all state and enforces all rules inside the database:
  - **Auth**: emailed sign-in links, and a trigger that rejects non-UC addresses.
  - **Postgres**: 8 tables, each with **row-level security**, plus triggers that fill in names, cap listings and block self-messaging.
  - **Realtime**: chat delivery.
  - **Storage**: the `listing-photos` bucket.
  - **pg_cron**: the nightly 30-day chat deletion.
- **Three Next.js API routes** run on Vercel serverless functions. They keep third-party keys and rate limits off the client: `/api/geocode` calls OpenStreetMap Nominatim; `/api/places/search` and `/api/places/rating` call Google Places API (New).
- **Phone app:** an Expo / React Native app that shares the web app's logic through a sync script. It still runs on local demo data; moving it onto Supabase is T20.

Deployment model. There is one Vercel project (root `uc-sublease/`, live at https://roost-uc-sublease.vercel.app) and one Supabase project (`roost-uc-sublease`, us-east-2, Free plan, Pro before launch). Schema changes are SQL migrations `001` to `007`, applied in order. Secrets live only in Vercel and Supabase settings (`GOOGLE_PLACES_API_KEY`, SMTP credentials); the shipped JavaScript carries only the publishable key.

Data flow.

- Sign-in: the browser asks Supabase Auth to email a link. The `handle_new_user` trigger rejects non-UC addresses before any email is sent and creates the profile. The link returns a JWT session.
- Reads and writes go from the browser to Supabase (PostgREST over HTTPS, with the JWT). RLS decides which rows each user sees; BEFORE INSERT triggers set server-owned fields.
- Posting: the address goes through `/api/geocode` to coordinates. Photos are compressed in the browser and uploaded to the host's own Storage folder. Then the listing row is inserted.
- Search, ranking and filtering run in the browser over the active listings RLS returns (no server search index at campus scale, A-01).
- Chat: rows are inserted into `threads` and `messages`, and Realtime pushes them to the other participant. `share_contact` and `get_contact` (SECURITY DEFINER functions) control email disclosure.

Diagrams. The prototype has no new diagrams. The D0 to D2 figures stay valid for the parts mapped below; they are linked, not duplicated: D0_High_Level_Design.pdf (block diagram, Data Flows 1 to 4), D1_Detailed_Design.pdf, D2_Detailed_Design.pdf and D2_figures/.

2.1 Design components mapped to the prototype

D0/D2 element	In the prototype	Status
C1 Web Client	Next.js app `uc-sublease/app/*` + Expo app `roost-mobile/`	Built (web); phone on local data
C2 Identity, G1 Auth	Supabase Auth email link; `handle_new_user` trigger; `email_allowlist`; JWT checked by PostgREST and `requireUser()`	Built
C3 Listing	`listings` table, RLS, `listings_before_insert` trigger; statuses active, rented, removed	Built (no hold state yet)
C4 Compliance Screener	D2 Algorithm D re-hosted as a database trigger (§3.6)	Designed, not built
C5 Search & Match	`filterListings()` and `rankByFit()` in the browser (D1 GiST index not used)	Built
C6 Messaging	`threads`, `messages`, Realtime, `share_contact`, `get_contact`, `mark_thread_read`	Built
C7 Approval	Withdrawn; replaced by `landlord_consent` CHECK and the fee split	Withdrawn
C8 Notification, E1	Supabase Auth email only (custom SMTP pending, T19); no other email	Partly built
G2 Configuration	Vercel/Supabase environment variables; `lib/config.ts`; migrations	Built
G3 Logging and audit	Supabase and Vercel platform logs; `reports` table; no staff audit log yet (T23)	Partly built
E3 Managed SQL	Supabase Postgres	Built
New: E4 Google Maps Platform, E5 OpenStreetMap	Places API (New), Maps JS; Nominatim geocoding and Leaflet tiles	Built

3. Interfaces and Data Formats

3.1 Conventions

- **Types:** dates are ISO `yyyy-mm-dd`; timestamps are `timestamptz` (UTC); rent and fees are integer US dollars; user ids are UUIDs; listing ids are text (UUID strings).
- **Authentication:** every Supabase call carries `Authorization: Bearer <Supabase JWT>` and `apikey: <publishable key>`. Calls to the three API routes carry the same Bearer token.
- **Supabase (PostgREST) error behavior:**
  - Invalid or expired JWT: **401**.
  - An RLS-filtered SELECT returns **200 with no rows** (it never reveals that the row exists).
  - An insert or update that RLS refuses: **403**, code `42501`.
  - A CHECK violation: **400**, code `23514`.
  - A unique violation: **409**, code `23505`.
  - A trigger `raise exception`: **400**, code `P0001`, with its message.
  - The client maps each case to one message from `lib/copy.ts`.
- **API route errors:** `{"error": "<code>"}`, where the code is one of `sign_in_required` (401), `bad_id` (400), `slow_down` (429), `not_found` (404), `lookup_failed` (502) or `not_configured` (503).
- **Security headers** (`next.config.ts`): `X-Frame-Options: DENY`, `frame-ancestors 'none'`, `X-Content-Type-Options: nosniff`, a strict `Referrer-Policy`, a `Permissions-Policy` that disables camera, microphone, location and payments, and `Cross-Origin-Opener-Policy`.

3.2 Component boundaries

Boundary	Inputs → outputs	Format	Protocol	Error behavior
User ↔ web client (I1)	Taps, swipes, keys, forms → pages, toasts, live counts	HTML, ARIA	HTTPS (browser)	Field errors shown inline from `lib/copy.ts`; motion off under reduced-motion; draft kept on the post form
Web client → Supabase Auth (I2)	`signInWithOtp({email})` (link); `signInWithPassword` (review account only) → session JWT	JSON	HTTPS	Non-UC address: trigger error "Only UC email addresses…", no email sent. Expired or used link: no session. Rate limit: 429
Web client ↔ Supabase DB (I3 to I6, I15)	Table reads and writes, RPC → rows	JSON	PostgREST over HTTPS	§3.1 codes; RLS hides others' rows; triggers override client-set names

Boundary	Inputs → outputs	Format	Protocol	Error behavior
Web client ↔ Realtime	Subscribe to `messages`, `threads` → inserted rows	JSON	WebSocket (TLS)	Only rows the user may read are delivered; on reconnect the client re-fetches the thread
Web client → Storage	`upload(<uid>/<file>)` JPEG/WebP/PNG ≤ 2 MB → public URL	Binary	HTTPS	Another user's folder: 403; type or size refused: 400; the listing is saved without the failed photo
Web client → `/api/*`	Query string + Bearer → JSON (§3.3)	JSON	HTTPS GET	401, 400, 429 (per-user, in memory), 502 upstream, 503 key missing
`/api/*` → E4/E5	Nominatim search; Places Text Search; Place Details (field mask)	JSON	HTTPS	Upstream failure → `lookup_failed`; ratings never cached (`no-store`); geocode cached 24 h
Supabase Auth → SMTP → UC Mail (I13, I14)	Sign-in link email	RFC 5322 MIME	SMTP over TLS	Built-in sender reaches only team members until custom SMTP (T19); bounces are visible in the SMTP provider
pg_cron → DB	`roost-delete-old-chats` at 03:17 UTC	SQL	In-database	A failed run is logged in `cron.job_run_details`; the next night catches up
Phone app ↔ device store	Listings, shortlist, chats (demo)	JSON	AsyncStorage	Local only, to be replaced by the Supabase data layer (T20)

3.3 Endpoint table

Next.js API routes (Vercel, Bearer token required):

Method and path	Request	Success	Errors	Limit	Req
GET /api/geocode	`q` 4 to 120 chars	200 `{results:[{label, area, lat, lng}]}` (at most 5; `[]` if q is short)	401; 429; 200 with `error:"lookup_failed"`	30/min/user	REQ-17, CON-09
GET /api/places/search	`q` 3 to 80 chars	200 `{results:[{id, name, address}]}`	401; 429; 502; 503	20/min/user	REQ-27
GET /api/places/rating	`id` = Place ID `[A-Za-z0-9_-]{10,200}`	200 `{name, rating, count, url}`	400 bad_id; 401; 404; 429; 502; 503	60/min/user	REQ-27, CON-09

Supabase data API (PostgREST `/rest/v1`, Bearer token required, RLS on every row):

Operation	Who may	Request / result	Errors	Req
SELECT listings	Signed in	Active listings, plus the caller's own in any status	401 (anon)	REQ-03, REQ-07, REQ-10
INSERT listings	Owner = caller, not sample	Listing row (§3.5) → row; trigger sets host_name and confirmed_at	400 (23514 checks; P0001 "20 active listings"); 403	REQ-04, REQ-06, REQ-20
UPDATE / DELETE listings	Owner	status rented, confirmed_at, fields	403	REQ-10
SELECT, UPDATE profiles	Self only; columns display_name, prefs, prefs_asked, onboarded	prefs JSON (§3.4)	403 for other columns	REQ-19, NFR-02
SELECT, INSERT, DELETE shortlist	Self	`{user_id, listing_id}`	409 duplicate	REQ-24
INSERT threads	Seeker = caller	`{listing_id}` → thread; trigger sets host, names, shared flags	P0001 "no longer available" / "your own listing"; 409 one thread per pair	REQ-08
SELECT, INSERT messages	Thread participants; `kind='user'`	`{thread_id, body 1–2000}`	403; 400	REQ-08, REQ-22
RPC share_contact(p_thread)	Participant	Sets the caller's shared flag and posts a system message	"Chat not found."	REQ-21
RPC get_contact(p_thread)	Participant	The other person's email, or null until they share	—	REQ-21, NFR-02
RPC mark_thread_read(p_thread)	Participant	Updates the caller's read_at	—	REQ-15
INSERT, SELECT reports	Reporter = caller	`{listing_id, reason 3–200}`	409 one per listing; 400	REQ-23
SELECT property_groups	Signed in	12 quick-pick companies `{name, kind, google_place_id}`	401	REQ-27
Storage `listing-photos`	Owner folder `<uid>/` write; public read	JPEG/WebP/PNG ≤ 2 MB	403; 400	REQ-25

3.4 Data formats

```
// profiles.prefs and listings.household (roommate ratings; household only on room listings)
{ "cleanliness": 4, "noise": 2, "social": 2, "guests": 1, "sleep": 3, "approachability": 4,
  "smokeFree": true, "cats": false, "dogs": true }

// Listing insert (client → PostgREST); server sets id, host_name, host_detail, confirmed_at, created_at
{ "owner_id": "<auth.uid()>", "title": "Sunny room near campus", "address": "…", "area": "Clifton",
  "lat": 39.13, "lng": -84.52, "rent": 650, "utilities_included": false, "utilities_estimate": 60,
  "start_date": "2027-01-04", "end_date": "2027-04-30", "bedrooms": 3, "furnished": true,
  "unit_type": "room", "transfer_type": "sublease", "amenities": ["wifi", "laundry"],
  "sublet_fee": 150, "fee_split_pct": 50, "landlord_consent": true,
  "google_place_id": "ChIJP3LeqIezQYgRs8U5H0Yrlxo", "photos": ["<uid>/a.jpg"], "thumbs": ["<uid>/a_t.jpg"] }
```

3.5 Database schema (Supabase Postgres, `public` schema)

Table	Key columns	Constraints that enforce requirements
profiles	id PK → auth.users, email, display_name, prefs JSONB, prefs_asked, onboarded	Created by trigger only; update limited to 4 columns; RLS self-only (NFR-02, REQ-19)
listings	id PK, owner_id (null = sample), is_sample, title, address, lat, lng, rent, dates, bedrooms, furnished, unit_type, transfer_type, amenities[], sublet_fee, fee_split_pct, household, landlord_consent, photos[], thumbs[], google_place_id, status, confirmed_at	CHECK end_date > start_date; rent 1 to 20000; title 3 to 120; ≤ 8 photos; `landlord_consent` must be true; status ∈ {active, rented, removed}; trigger caps 20 active per owner (REQ-04, REQ-20, NFR-05)
shortlist	(user_id, listing_id) PK	RLS self (REQ-24)
threads	id PK, listing_id, seeker_id, host_id, names, seeker_shared, host_shared, read_at × 2	UQ (listing_id, seeker_id); no UPDATE/DELETE grant; trigger blocks inactive and own listings (REQ-08, NFR-06)
messages	id PK, thread_id, sender_id, kind ∈ {user, system, sample_host}, body 1 to 2000	Insert only as self with kind 'user' in own thread; cron deletes after 30 days (REQ-22)
reports	id PK, listing_id, reporter_id, reason 3 to 200	UQ (listing_id, reporter_id) (REQ-23)
email_allowlist	email PK	RLS on, no policies: invisible to the API (REQ-01)
property_groups	id PK, name, kind, google_place_id UQ	Read-only to signed-in users (REQ-27)

3.6 Designed, not built: Fair Housing screening (REQ-05, REQ-09, NFR-04)

- **Algorithm:** D2 Algorithm D, unchanged: normalize, fold common substitutions, match whole words, flag mixed-script tokens.
- **Where it runs:** instead of an in-process C4 module, it runs inside the `listings` BEFORE INSERT/UPDATE trigger. It reads a new staff-only `phrase_list` table, with the version stored on the row.
- **What changes:**
  - Status gains `held`.
  - The RLS read policy already hides non-active rows from other users (AC-01.3).
  - New columns: `flagged_phrases text[]`, `reviewed_by`, `reviewed_at`.
- **Fail-closed by construction:** if the trigger errors, the insert fails and no active row can exist (NFR-04).
- **Staff release or removal** (REQ-09) goes through a staff-only RPC that writes `reviewed_by` and `reviewed_at`. Tasks T4, T8 and T23.

4. Technology Stack

Layer	Choice	Justification (requirement or constraint IDs)
Web front end	Next.js 16.4 (App Router), React 19, TypeScript 5, Tailwind 4, shadcn button, Base UI dialog	Native elements and a reduced-motion setting: NFR-01. Thumbnails and lazy loading: NFR-08. One language for web and phone: CON-07
Phone app	Expo SDK 57 / React Native, Expo Router; shared `lib/` via `npm run sync`	Most seekers browse on phones (US-02); one TypeScript codebase: CON-07; Expo Go needs no store release (§7.3 of the RD)
Data and rules	Supabase Postgres with RLS, CHECK constraints, triggers, SECURITY DEFINER functions	Rules are enforced server-side, so the client cannot bypass them: REQ-03, REQ-04, REQ-20, NFR-02, NFR-03, NFR-06
Authentication	Supabase Auth email link plus the domain trigger; one password review account	REQ-01, REQ-02; no SSO available: CON-06
Real-time	Supabase Realtime on `messages` and `threads`	REQ-08, REQ-15

Layer	Choice	Justification (requirement or constraint IDs)
File storage	Supabase Storage `listing-photos` (public read, owner-folder write, 2 MB, images only); browser compression to 1,600 px and 480 px	REQ-25; keeps storage within the free tier: CON-01
Scheduled jobs	pg_cron	REQ-22, with no extra service: CON-01, CON-05
Server routes	Next.js route handlers on Vercel; `requireUser()`; in-memory per-user limiter	Keys stay off the client: NFR-12; abuse limits: NFR-05; third-party terms: CON-09
Hosting	Vercel (Hobby) + Supabase Free, then Supabase Pro before launch	Managed only: CON-05; $0 now and about $25 per month at launch: CON-01; no idle pause: NFR-11
Maps and geocoding	Leaflet + OpenStreetMap tiles (Browse map); Google Maps JS (stop map); Nominatim (geocode, cached 24 h)	REQ-17, REQ-18; no key for most maps: CON-01; 1 request per second policy: CON-09
Ratings	Google Places API (New): Text Search + Place Details (rating fields only)	REQ-27; free monthly calls under CON-01; Place-ID-only storage: CON-09
Email	Supabase Auth through custom SMTP (a Gmail app password for the class test, a domain sender later)	NFR-09; deliverability to UC Mail: CON-08
Verification	`supabase/tests/rls-checks.sql` (28 checks, exists); planned: GitHub Actions, Vitest (ranking, distance and fee math), Playwright end-to-end, axe-core, Lighthouse, an uptime monitor	NFR-03; REQ tests TC-0nn; NFR-01, NFR-08, NFR-11 (T25)

Removed since v1.0 (§7): Express, Render, jose, the Postgres outbox and relay, the server search index (GiST), and the C8 notification queue.

5. Known Constraints

ID	Constraint (RD §8)	Rules out	Forces
CON-01	$0 now, about $25 per month at launch	Paid search or moderation services, a message broker, extra servers	Free tiers; client-side search; pg_cron; Google quota caps and a budget alert; Supabase Pro only at launch
CON-02	Contact privacy; row-level access	Showing emails in listings or chats; client-side-only access checks	RLS on every table; `get_contact` only after Share; trigger-set names
CON-03	Fair Housing Act / Ohio R.C. 4112	Protected-characteristic filters; unscreened publishing; roommate questions that mirror protected classes	Screening trigger (§3.6); report button (REQ-23); fairness review of roommate wording (GQ-12)
CON-04	WCAG 2.1 AA	Colour-only status; gestures as the only control	Buttons and arrow keys as alternatives to swiping; icon tags with text labels; reduced motion; axe-core gate (T17)
CON-05	Managed services only	Self-hosted servers or SMTP	Vercel + Supabase; config in environment variables and migrations
CON-06	No UC SSO	SAML/Shibboleth	UC-domain email link plus the trigger; allowlist for staff
CON-07	Three students, Apr 16, 2027	Separate native codebases; manager onboarding	Expo with shared `lib/`; the MoSCoW cut in RD §7
CON-08	Authenticated email to UC Mail	Supabase's built-in sender for real users	Custom SMTP (T19); Auth rate limits; CAPTCHA (T22)
CON-09	Google and OpenStreetMap terms	Caching ratings; storing Places content other than Place IDs; unthrottled Nominatim	`no-store` rating route; Place-ID column only; server-side geocode with a User-Agent and 24 h cache

6. Traceability Matrix

Status: **In progress** = built in the prototype at `71d351b`, with Week 10 verification pending. **Designed** = specified (D0 to D2 or §3.6) but not built. **Planned** = not yet designed. Test IDs are reserved for the Week 10 plan: REQ-nn → TC-0nn, NFR-nn → TC-1nn, CON-nn → TC-2nn (GQ-10).

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
REQ-01	UC email or allowlist only, rejected before any email	D0 C2, D-02; D2 Algorithm C (EC-C2-05) → prototype `handle_new_user` trigger, `email_allowlist`	In progress	TC-001
REQ-02	One-time link sign-in; expired or used link fails	D0 I2, D-09; D2 Fig. 8 → Supabase Auth link; review-account password	In progress	TC-002
REQ-03	Anonymous requests refused	D2 G1 (I18) → RLS on all tables; `requireUser()`	In progress	TC-003
REQ-04	Invalid listing rejected, nothing written	D1 §5 POST /listings; D2 EC-C3-01 → `listings` CHECKs	In progress	TC-004

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
REQ-05	Screen listings; hold flagged ones	D0 C4, I7; D2 Algorithm D → §3.6 trigger	Designed (list content: T4)	TC-005
REQ-06	Listing visible to others within 60 s	D1 Algorithm B (outbox removed) → direct insert + RLS read policy	In progress	TC-006
REQ-07	Whole-range date match, term presets	D0 D-06; D1 Algorithm A → `filterListings()`, `TERMS`	In progress	TC-007
REQ-08	Private real-time chat, one thread per listing per seeker	D0 C6, I5; D2 Part 2.5, Fig. 3 → `threads`, `messages`, Realtime	In progress	TC-008
REQ-09	Staff release or remove held listings, audited	D1 §5 review endpoints; D2 R-10 → §3.6 staff RPC	Designed	TC-009
REQ-10	Host sees own listings; re-confirm or mark rented	D2 GET /listings/mine (Part 2.8) → `/mine`, RLS owner read	In progress	TC-010
REQ-15	Real-time unread count	D2 C6 read marks → `mark_thread_read`, notification bell	In progress	TC-015
REQ-16	Rent, bedroom, furnished filters	D1 Algorithm A (maxRentCents) → `filterListings()`	In progress	TC-016
REQ-17	Distance from geocoded address; mile filter	D0 C5 (search) → `/api/geocode`, `lib/geo.ts`, `DISTANCE_OPTIONS`	In progress	TC-017
REQ-18	Nearest shuttle and bus stop; stop map	D0 C1/C5 → `lib/transit.ts` (53 real shuttle stops; sample bus stops), stop-map sheet	In progress	TC-018
REQ-19	Roommate preferences; fit ranking; Show all	D0 C5 (ranking) → `lib/roommates.ts`, `rankByFit()`, MATCH_MIN 55	In progress	TC-019
REQ-20	Landlord consent required; fee split	D0 C3 (replaces C7, D-11) → `landlord_consent` CHECK, `sublet_fee`, `fee_split_pct`	In progress	TC-020
REQ-21	Email revealed only after Share	D2 R-05, NFR-02 → `share_contact`, `get_contact`	In progress	TC-021
REQ-22	Messages deleted after 30 days	D2 C6 data model → pg_cron `roost-delete-old-chats`	In progress	TC-022
REQ-23	Report a listing once; staff remove	D2 GQ-02 (no report feature) → `reports` table, report dialog, SQL removal	In progress	TC-023
REQ-24	Save and compare 2 or 3 listings	D0 C1 → `shortlist`, compare table	In progress	TC-024
REQ-25	Up to 8 photos, compressed, owner folder	D0 C1, C3 → `lib/images.ts`, Storage policies	In progress (web)	TC-025
REQ-26	Swipe deck, list and map with shared filters	D0 C1 (D-08) → `swipe-deck.tsx`, `/list`, Leaflet map	In progress	TC-026
REQ-27	Live Google rating of the property company	D0 C1 → `/api/places/*`, `property_groups`	In progress	TC-027
REQ-28	"May be taken" after 10 days unconfirmed	D2 GQ-03 (listing lifecycle) → `confirmed_at`, `freshness()`	In progress	TC-028
REQ-29	LLM suggested questions with a spending cap	**ORPHANED**: no D0 to D2 element; design waits for the advisor's policy and cap (GQ-15), then T27	Planned	TC-029
NFR-01	WCAG 2.1 AA, 0 axe violations, screen-reader pass	D0 I1; D2 Part 2.2 → reduced-motion support; full pass is T17	In progress	TC-101
NFR-02	No email exposed without Share	D2 R-05, EC-G3-03 → RLS, `get_contact`	In progress	TC-102
NFR-03	RLS everywhere; anonymous refused; 28 checks pass	D2 G1 → RLS policies, `rls-checks.sql`	In progress	TC-103
NFR-04	Screening failure never publishes	D0 I7 fail-closed; D2 Fig. 7 → §3.6 trigger	Designed	TC-104
NFR-05	Lookup limits; 20-listing cap; Auth limits; CAPTCHA	D2 Algorithm C step 3 → `allow()` limits, listing trigger; CAPTCHA is T22	In progress	TC-105
NFR-06	Server-set names; no partial rows	D2 NFR-06, R-06 → BEFORE INSERT triggers, single-statement inserts	In progress	TC-106
NFR-08	LCP ≤ 2.5 s p75; filter update ≤ 200 ms	D1 Algorithm A (complexity) → thumbnails, client-side filtering	In progress	TC-108
NFR-09	95% of sign-in emails within 60 s, not junk	D0 E1, I13, I14 → custom SMTP (T19)	Designed	TC-109

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
NFR-10	4 of 5 testers find and message within 3 min	D2 Part 2.2 screen table → class test (T18)	In progress	TC-110
NFR-11	≥ 99% uptime over 7 days; no idle pause	D2 G3 /healthz → Vercel + Supabase Pro (T26)	Designed	TC-111
NFR-12	No secrets in the bundle; Maps key restricted	D2 G2 (secrets in environment) → server-only Places key; restricted browser key is T24	In progress	TC-112
CON-01	$0 now, about $25 per month at launch	D0 D-01 (budget) → free tiers, Google quota caps	In progress	TC-201 (cost analysis)
CON-02	Contact privacy, row-level access	D0 Part 6 → RLS, `get_contact`	In progress	TC-202 (inspection)
CON-03	Fair Housing Act / Ohio R.C. 4112	D0 C4, D-04 → §3.6, reports, wording review	Designed	TC-203
CON-04	WCAG 2.1 AA	D0 I1, D-08 → C1 alternatives to gestures	In progress	TC-204
CON-05	Managed services only	D0 E3, D-01 → Vercel + Supabase	In progress	TC-205 (inspection)
CON-06	No UC SSO	D0 D-02, D-09 → email link + trigger	In progress	TC-206
CON-07	Team and schedule	A8 plan → Tasklist.md v2	In progress	TC-207 (tracking)
CON-08	Authenticated email to UC Mail	D0 E2, I14 → custom SMTP, CAPTCHA (T19, T22)	Designed	TC-208
CON-09	Google and OSM terms	D0 C1 (new external E4/E5) → `no-store` routes, Place-ID-only column	In progress	TC-209 (inspection)

REQ-11 to REQ-14 and NFR-07 are withdrawn (RD v2.0) and have no rows. **Orphaned: 1 (REQ-29)**, explained above.

7. Design Changes Since the Week 7 Review (D2, 2026-10-14)

Change	Why	Effect
Custom Node/Express modular monolith on Render → Supabase backend-as-a-service + Next.js on Vercel	Advisor meeting 1 asked to move to Supabase; rules enforced in the database cover D2's G1, C2, C3 and C6 with far less code (CON-07)	D-01, D-05 (outbox), D-03 (own JWT) superseded; §2.1 mapping
Property-manager approval (C7, US-03, REQ-11 to REQ-14) withdrawn	Team decision built on the advisor's meeting-1 request for landlord consent and a fee split; no real manager to test with (RK6)	REQ-20 added; UC-02, AC-02.x withdrawn
C4 screener not built; re-hosted as a trigger	Prototype time went to the advisor's first priorities (distance, form, preferences)	REQ-05, NFR-04 stay Must; §3.6; T4, T8
Server search index (D1 GiST) → client-side filtering and fit ranking	A few hundred listings (A-01); ranking needs per-user preferences in the browser	D1 Algorithm A superseded
Added distance, map, stops, roommate fit, fee split, Google ratings, compare, reports, photo upload, swipe deck	Advisor meeting 1 requests	REQ-17 to REQ-28
`@uc.edu` admitted, not only `@mail.uc.edu`	Advisor meeting 1: faculty, staff and grad students	REQ-01
Chat privacy: TLS + at-rest encryption + 30-day deletion, no E2EE	Advisor decision 2026-10-25	REQ-22; RD §7.3 Won't
Error-code rule (D2 R-08) → PostgREST codes (§3.1)	No custom API layer to apply R-08	§3.1

8. Advisor Review Record

Date	Form	What was discussed	Changes it produced
Meeting 1 (date: _____, before 2026-10-21)	Meeting	Replace neighborhoods with distance from UC and a map; nearest shuttle and bus stop; a fuller listing form; sublet fee after landlord consent with a split; icon amenities; property Google reviews; open to faculty, staff and grad students; roommate rating scales and fit ranking with Show all; photo compression; 30-day chats; move to Supabase; a weekly checklist	REQ-01 (`@uc.edu`), REQ-17 to REQ-20, REQ-22, REQ-24 to REQ-27; US-06 to US-08; the backend change in §7

Date	Form	What was discussed	Changes it produced
2026-10-21	Meeting (demo and the prep questions in roost-uc-sublease `docs/archive/`)	Demo of the prototype; backend ownership; sign-in without SSO; the chat privacy model; transit data; distance meaning; roommate wording and fairness; LLM cap; landlord-consent proof; test plan; timeline. Priorities agreed: distance search, listing form, user preferences, then encryption and image size	Priority order in RD §7.2; open questions GQ-11 to GQ-15
2026-10-22	Written progress update (roost `docs/roost-advisor-update-2026-10-22.pdf`)	Status after the move to Supabase; the email-setup blocker; next steps 1 to 8	Tasklist T19 to T27
2026-10-25	Decision	Chat privacy: encryption in transit and at rest with 30-day retention is enough; no end-to-end encryption; server access limited to authorized staff	REQ-22; E2EE on the Won't list; GQ-13 narrowed to UC's retention rule
_____	Review of RD v2.0 and TS v2.0 drafts (bring the §6 matrix)	_____	*Log as v2.1*

Suggested agenda for the draft review: (1) Are REQ-05 and NFR-04 (screening) right as Must even though they are not built yet? (2) The CON-01 budget change ($7 → about $25 per month at launch). (3) GQ-11 to GQ-15 and GQ-18. (4) The orphaned REQ-29.